

Kevin Jiang

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Education

University of Waterloo

Bachelor of Software Engineering (Honours)

Sep 2024 – Apr 2029

Experience

Software Engineer - Geospatial AI Intern, Epoch Blue | London, UK

Sept 2026 - Dec 2026 | [About](#)

Full-Stack Developer, Royal Canadian Air Force (RCAF) | Ottawa, ON

Jan 2026 - Apr 2026 | [About](#)

- Built and **deployed two web applications** for RCAF operations, streamlining workflows and improving access to data
- Implemented RBAC in AWS Cognito whilst incorporating **domain-driven design**, enhancing security with modularity
- Designed and tested **50+ RESTful API** endpoints and **15+ schemas** in PostgreSQL powering core app features
- Built **10+ webpages** and components using Vue.js, collaborating with UI/UX designers for user-centric design
- Reviewed **25+ pull requests**, improving code quality, correctness, and data consistency

Data Engineering Research Assistant, University of Waterloo | Waterloo, ON

May 2025 - Aug 2025

- Aggregated **10.7 million** geospatial data points from **22 sources** into PostgreSQL databases using ArcGIS and Python
- Built and documented **5 modules** in R/Python for data analysis use with the **Climate Risk Research Group**
- Developed **6 web dashboards** using React, R Shiny, and RMarkdown to visualize assorted flood data
- Prototyped approaches integrating novel data and techniques into **flood loss models**, evaluating impact on accuracy

Software Engineer Intern, ReliablyME | Toronto, ON

Jan 2025 - Apr 2025 | [Github](#)

- Black-box tested **2400+ lines** of RESTful API code used in AI-driven commitment tracker startup, improving reliability
- **Implemented features** allowing users to backdate or update their dates of commitment completion

Projects

1st Place, Factual @ SOA Research Challenge | 68 teams | R, Python, NumPy

Mar 2025 | [Paper](#) | [GitHub](#)

- **Awarded \$10000** for designing a national flood insurance program with policy experts using **layered risk modeling**
- Built **machine learning pipelines** using Google's YDF for EDA, parameter tuning, and model validation in R/Python
- Featured on [SOA podcast](#), [The Actuary](#), ImpACT Conference, and the [UWaterloo Math Department](#)

2nd Place, UTRAHacks 2025 | 249 participants | C++, Arduino, Robotics

Jan 2025 | [GitHub](#)

- Built an Arduino-based robotic control systems for a robotics navigation and path-finding competition
- Programmed **C++ state machine algorithms** to complete path-finding challenges

3rd Place, Debug Debacle @ McHacks 2025 | 471 participants | Python, React.js, SocketIO

Jan 2025 | [GitHub](#) | [Link](#)

- Published **full-stack multiplayer game** with OAuth and Vercel where players race to fix AI-generated bugged code
- Wrote **code validation scaffolds** to test if player submissions correctly solved GumloopAI AI-generated bugs

Top 3 Finalist, 2025 ASNA Hackathon | 37 teams | Python, NumPy, Pandas, XGBoost

Jan 2025 | [Homepage](#) | [Github](#)

- Developed **XGBoost and YDF machine learning models** with Python and Pandas for claim prediction
- Achieved **93.6% accuracy** on projecting car insurance claims for actuarial science competition

Firmware Member, Waterloop | C++, STM32, CAN bus protocol

Sep 2024 - Present | [Github](#)

- Integrated new Raspberry Pi as central control unit of the design team's machine via **CAN protocol**
- Developed and documented **STM32 scripts** using C++ to verify proper operation of the CAN network

Skills and Qualifications

- **Languages:** Python, JavaScript, Typescript, C++, C, R, SQL, HTML/CSS, Bash, Java, Scala
- **Libraries/Frameworks:** Vue.js, React.js, Axios, Express.js, Pinia, SQLite, STM32, CASBIN, Geopandas, Pandas, Shiny, Tkinter, Flask, NumPy, Tensorflow, PyTorch, RMarkdown, Bootstrap, Datatables, Quill.js, Vuetify, Vueflow, Leaflet
- **Tools/Technologies:** Node.js, Docker, Google Cloud Platform, AWS, PostgreSQL, MongoDB, MSSQL Server, Sequelize, Apache, Nginx, Debian, Ubuntu, Raspbian, Hazus, ARCGIS, QGIS